

Sequences, Series & Complex Numbers



Arithmetic sequences

- 1 Find the 12th term of the arithmetic sequence 5, 9, 13, ...
 - 2 The 4th term of an arithmetic sequence is 18, and the 9th term is 38. Find the first term and the common difference.
 - 3 Find the sum of the first 20 terms of the arithmetic sequence 7, 11, 15, ...
 - 4 Find the number of terms in the arithmetic sequence 6, 10, 14, ..., 150.
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Geometric sequences

- 5 Find the 7th term of the geometric sequence 3, 6, 12, ...
 - 6 In a geometric sequence with all positive terms, the 3rd term is 20 and the 5th term is 80. Find r and a_1 .
 - 7 Given that $x - 1$, $x + 1$, and $3x - 1$ are consecutive terms of a geometric sequence, find the possible value(s) of x .
 - 8 Find the sum of the first 6 terms of the geometric sequence 5, 15, 45, ...
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Series and sigma notation

- 9 Evaluate $\sum_{k=1}^5 (3k - 2)$.
 - 10 Write the series $3 + 7 + 11 + \dots + 39$ in sigma notation, then evaluate it.
 - 11 Evaluate $\sum_{n=1}^6 2^n$.
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Geometric series

- 12 Find the sum of the first 8 terms of the geometric sequence 2, 6, 18, ...
- 13 Find the sum to infinity of $16 - 8 + 4 - 2 + \dots$

- 14 Express the repeating decimal $0.4\overline{5}$ (that is, $0.45555\dots$) as a geometric series and find its exact fractional value.
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Complex number arithmetic

- 15 Given $z_1 = 5 - 2i$ and $z_2 = -3 + 4i$, find $z_1 + z_2$ and $z_1 - z_2$.
- 16 Multiply $(3 + 2i)(4 - 5i)$.
- 17 Simplify i^{23} , using $i^2 = -1$.
- 18 Simplify $(2 - 3i) - (5 + i) + (1 - 4i)$.
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Complex conjugates and division

- 19 Find the conjugate of $z = 6 + 7i$, and compute $z\bar{z}$.
- 20 Simplify $\frac{4 + 3i}{1 - 2i}$ by multiplying by the conjugate of the denominator.
- 21 Simplify $\frac{5}{3 + i}$ by multiplying by the conjugate of the denominator.
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Modulus of complex numbers

- 22 Find $|z|$ for $z = -8 + 15i$.
- 23 If $z_1 = 3 + 4i$ and $z_2 = 1 - i$, find $|z_1 z_2|$.
- 24 If $z = 4 - 3i$, find $|z|$ and $|2z|$.
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Quadratic equations with complex roots

- 25 Solve $x^2 + 4x + 13 = 0$, giving your answer in the form $a + bi$.
- 26 One root of a quadratic equation with real coefficients is $2 - 5i$. Find the other root, and write the equation in standard form.
- 27 Solve $2x^2 - 4x + 5 = 0$, giving your answer in the form $a + bi$.
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A well-designed word problem: a bouncing ball

- 28** This question has three parts. A ball is dropped from a height of 20 meters. Each time it bounces, it rises to 60% of its previous height. (a) Find the height of the ball's rebound after its 4th bounce. (b) Write an expression for the total vertical distance traveled by the ball as it bounces infinitely many times, and evaluate it. (c) After which bounce does the rebound height first fall below 1 meter?